Fortgeschrittene Analysis (ANA-F)

Semesterwoche 1

Trigonometrische Funktionen

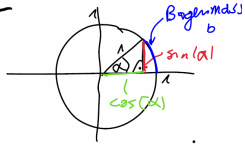
Ron Porath und Joachim Wirth

16. Februar 2026

Themen und Lernziele

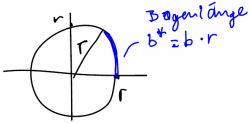
- Themen
  - Trigonometrische Funktionen:
    Sinus, Kosinus, Tangens, Cotangens
  - Umkehrfunktionen der trignometrischen Funktionen:
    Arkussinus, Arkuskosinus, Arkustangens, Arkuskotangens
  - Trigonometrische Gleichungen
- Lernziele
  - Sicherer Umgang mit den obengenannten Funktionen
  - Ableitungen der trigonometrischen Funktionen und ihrer Umkehrfunktionen berechnen
  - Lösen trigonometrischer Gleichungen

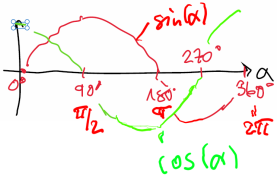
Trigonometrische Fkten



$2\pi \mathrel{\hat{=}} 360°$

$b \mathrel{\hat{=}} \alpha$

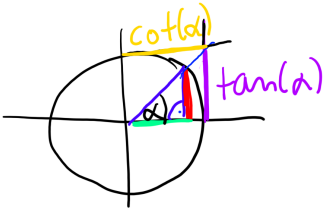




$\sin(x) = \sin(x + 2\pi) = \sin(x + k \cdot 2\pi)$

$\sin(x) = \cos(x - \frac{\pi}{2})$

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \qquad \cot(x) = \frac{\cos(x)}{\sin(x)}$$



$$\sin(2x) \overset{?}{=} \sin(x+x) = \sin(x)\cos(x) + \cos(x)\sin(x) = \underline{\underline{2\sin(x)\cos(x)}}$$

Additionstheoreme:

$$\sin(x_1 \pm x_2) = \sin(x_1)\cdot\cos(x_2) \pm \cos(x_1)\cdot\sin(x_2)$$

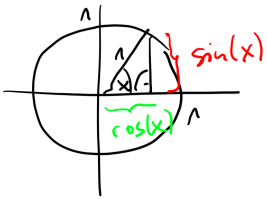
$$\cos(x_1 \pm x_2) = \cos(x_1)\cos(x_2) \mp \sin(x_1)\sin(x_2)$$

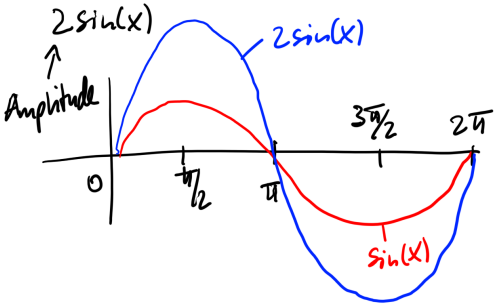
$$\cos(2x) = \cos(x)\cos(x) - \sin(x)\sin(x) = \underbrace{\cos^2(x)}_{1-\sin^2(x)} - \sin^2(x) = (1-\sin^2(x)) - \sin^2(x)$$

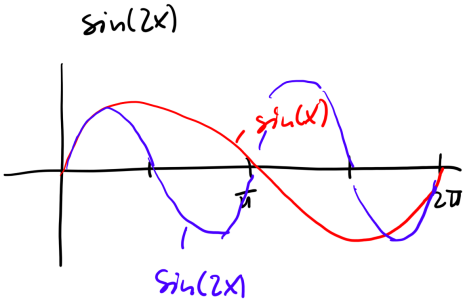
$$= \underline{\underline{1 - 2\sin^2(x)}}$$

<u>Trig. Pyth.</u>

$$\sin^2(x) + \cos^2(x) = 1$$

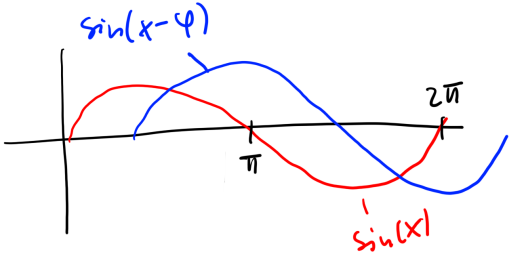






$\sin(0) \to \sin(0°)$

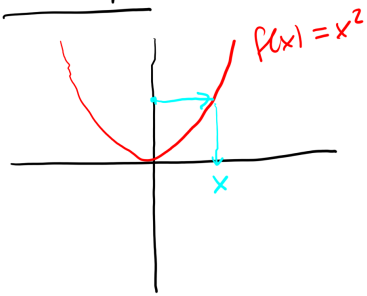
$\sin(45°) \to \sin(90°)$

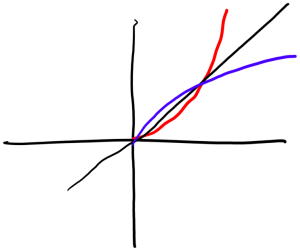


(Add. Th.)

$$y = \sin(x+\pi)$$
$$= \sin(x)\underbrace{\cos(\pi)}_{-1} + \cos(x)\underbrace{\sin(\pi)}_{0}$$
$$= \underline{\underline{-\sin(x)}}$$

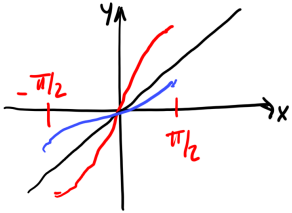
Umkehrfkt





1. $y = x^2 \rightarrow \sqrt{y} = x$ nach x auflösen

2. $y = \sqrt{x} =: f^{-1}(x)$ Variablen vertauschen



1. $f(x) = \sin(x) = y$

$\exists \sin^{-1}$ = Umkehrfkt.

$\sin^{-1}(\sin(x)) = \sin^{-1}(y)$

$x = \sin^{-1}(y) = \arcsin(y)$

2. $y = \arcsin(x)$

$$\arcsin(x) = -\frac{\pi}{6} \qquad x = ?$$

$$\sin(\arcsin(x)) = \sin\left(-\frac{\pi}{6}\right)$$

$$\underline{\underline{x = -0.5}}$$

Trig. Gleichungen

$$\sin(x) = \sqrt{1 - \sin^2(x)} \qquad x = ?$$

$$\sin^2(x) = \underbrace{1 - \sin^2(x)}_{\cos^2(x)}$$

$$\sin^2(x) = \cos^2(x)$$

$$\frac{\sin^2(x)}{\cos^2(x)} = 1$$

$$\frac{\sin^2(x)}{\cos^2(x)} = \frac{\sin(x)\sin(x)}{\cos(x)\cos(x)} = \tan^2(x) = 1$$

$$\tan(x) = \sqrt{1} = 1$$

$$\arctan(\tan(x)) = \arctan(1)$$

$$x = \arctan(1)$$

$$\underline{\underline{x = \pi/4}}$$

Trig. Gl.

$$\sqrt{\cos(x-1)} = \frac{1}{\sqrt{2}} \qquad x = ?$$

$$\cos(x-1) = \frac{1}{2}$$

$$\arccos(\cos(x-1)) = \arccos\left(\frac{1}{2}\right)$$

$$x-1 = \arccos\left(\frac{1}{2}\right)$$

$$x = 1 + \underbrace{\arccos\left(\frac{1}{2}\right)}$$

$$x = 1 + \pm \pi/3$$

$$x_1 = 1 + \frac{\pi}{3}$$

$$x_2 = 1 - \pi/3$$

Ableitung

Quotientenregel: $y = \frac{3x}{x^2} \Rightarrow y' = \frac{3 \cdot x^2 - 3x \cdot 2x}{x^4} = \frac{3x^2 - 6x^2}{x^4} = \underline{\underline{-\frac{3}{x^2}}}$

$$y = \frac{\sin(x)}{\cos(x)} \Rightarrow y' = \frac{\cos(x)\cos(x) + \sin(x)\sin(x)}{\cos^2(x)} = \underline{\underline{\frac{1}{\cos^2(x)}}}$$

$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$

$f(x) = \cos(x) \Rightarrow f'(x) = -\sin(x)$

Kettenregel: $y = (x+1)^2 \Rightarrow y' = 2(x+1) \cdot 1 = \underline{\underline{2(x+1)}}$

Wichtige Formeln

- Trigonometrischer Pythagoras

$$\sin^2(x) + \cos^2(x) = 1$$

- Additionstheoreme

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y)$$

$$\cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y)$$

- Tangens und Kotangens

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\cot(x) = \frac{\cos(x)}{\sin(x)} = \frac{1}{\tan(x)}$$

„Do It Yourself“

- **Theorie**

  Papula Band 1 III
  9, Seiten 243 – 258, Trigonometrische Funktionen
  10, Seiten 271 – 277, Arkusfunktionen
  10.5, Seiten 278 – 279, Trigonometrische Gleichungen

- **Übungen**

  Papula Band 1 III
  Übungsaufgaben zu den Abschnitten 9 und 10, Seiten 317 – 320
  Empfohlen: Aufgaben 1, 4, 5, 13 und 16
  Andere Aufgaben: Optional
  Lösungen: Seiten 756 – 761